

# RICHARD GUO

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## EDUCATION

### Columbia University

New York, NY

Bachelor of Science in Computer Science; GPA: 3.85/4.0

Expected May 2028

Coursework: Data Structures and Algorithms, Fundamentals of Computer Systems, Operating Systems, Computer Networks, Databases, Machine Learning, Competitive Programming, Probability Theory

## TECHNICAL SKILLS

**Languages:** Python, Java, Kotlin, TypeScript/JavaScript, C/C++, SQL

**Frameworks/Libraries:** React, React Native, Next.js, Flask, PyTorch, scikit-learn, OpenCV

**Tools/Platforms:** AWS (Lambda, DynamoDB, CDK), PostgreSQL, Docker, GitHub Actions, Git

## EXPERIENCE

### Amazon

Austin, TX

Software Engineering Intern

May 2026 – August 2026

- Built a rack visualization and auditing tool in React and TypeScript replacing **~4-hour manual audits** at fulfillment centers, deployed on live site data and validated across all 4 standard site layouts in a **1,000+ site fleet**.
- Designed and shipped the backend: 4 Kotlin Lambda APIs on DynamoDB handling placement edits, batched partial writes, and conflict rules keeping automated updates from overwriting human edits.
- Built auto-placement matching unracked devices to documented rack layouts by hostname, reducing per-rack placement time from **~5 minutes to under a second**.
- Shipped a placement rule engine (slot overlap, rack overrun, power-phase mismatch) reused by both read and write paths: zero false positives across 1,005 live devices at P95 1.7s against a 2s SLA.
- Built a nightly Python job mapping 260+ devices per site to their server rooms, reconciling two upstream inventory systems that track different hardware types.

### SetPoint.fit

New York, NY

Software Engineering Intern

January 2026 – May 2026

- Owned the core court-discovery experience (interactive map, searchable list, detail views) in React Native and TypeScript for a live iOS app with 300+ players and 50+ five-star App Store ratings.
- Rendered 170+ NYC courts with real-time busyness and distance ranking, integrating Flask geo-search APIs and Google Maps with synchronized map and list state.

### Creative Machines Lab at Columbia University

New York, NY

Computer Vision Research Assistant

September 2025 – December 2025

- Built a camera-projector AR guidance system for robot assembly, implementing an OpenCV/CoTracker vision pipeline holding 20 fps at 4K with calibration mapping tracked points to projector coordinates.

### Pennsylvania State University

University Park, PA

Natural Language Processing Research Assistant

June 2025 – August 2025

- Fine-tuned small language models on 50M+ tokens in PyTorch using the TinyStories dataset, identifying hyperparameter configurations that cut wall-clock training time  $3\times$  at equivalent validation loss.
- Cut evaluation time 70% with a GPT-based automated grading pipeline scoring 1,000+ generated samples.

### Jane Street

New York, NY

Math and Programming Externship (AMP)

June 2024 – August 2024

- Developed Python trading bots implementing fair-price estimation and penny strategies against a simulated exchange, finishing Top 10 in the Jane Street Electronic Trading Competition.

## PROJECTS

### PreSnap: NFL Win Probability Pipeline | Python, PostgreSQL, scikit-learn, Docker, GitHub Actions    Sept. 2026

- Built a pipeline ingesting 15 seasons of NFL play-by-play (600K+ plays) into PostgreSQL, with point-in-time features and an automated CI invariant test guarding against future-data leakage.
- Trained a gradient-boosted win-probability model (0.151 Brier, benchmarked against nflfastR) and shipped it as a live Streamlit app, with calibration analysis surfacing systematic overconfidence in one-score fourth quarters.